Coordination and Analysis of GSO Satellite Networks

BR-SSD e-Learning Center

Summary:

1) How to Identify Satellite Networks and other Systems for which Coordination is Required ?

2) Several Interference Criteria utilized to evaluate compatibility between GSO satellite networks.

   - ✓ Trigger Arc
   - ✓ DT/T
   - ✓ C/I

3) New Coordination Criteria under Study

4) Possible methods to be used to facilitate coordination and sharing scenario between GSO.

5) How to optimize a filing to be submitted to ITU ?

Identification of Coordination Requirements

Appendix 5 indicates technical criteria utilized in every case, including:

- Regulatory provision containing form of coordination
- Sharing scenario associated to the case
- Frequency Band and Region
- Services
- Threshold/condition
- Calculation Method

See Table 5-1, Table 5-2 and Annex 1 to AP5

Coordination between GSO/GSO under provision 9.7

New !
WRC'12

✓ Criterion of Coordination Arc:

- To identify satellites with frequency overlap operating in the same direction inside the window of ± 7 , ± 8 , ± 12 or ± 16 degrees from nominal orbital longitud, depending on the frequency band, service and Region.

- Applicable to satellite networks in FSS (non plan)
  BSS (non plan)
  Meteorological-Satellite
  associated Space Operations

  in specific frequency bands (see AP5)

- Utilized by BR to identify coordination requirements.

Simple but Useful Exercise:

Approach Identifying Satellite Networks with which coordination may be required by using:

SNS-Online or SpaceQuery: SNS Online Link→ http://www.itu.int/sns/query_builder.html

QUERY BUILDER

To build a query with a combination of data items of your choice check "Non-Planned Services" or "Planned Services" from the radio buttons, specify general parameters (the type and name of the satellite, category, notifying administration , longitude range and frequency range) then select data items from the list. Once you have made your selection click "Submit" button.

◉ Non-Planned Services ○ Planned Services/SOF

Enter query title

General Parameters	
Satellite Type:	◉ Geo ○ Nongeo ○ Both
Satellite Name:	
Notifying Administration:	
Network Organization:	
Longitude from:	12 to: 28
Frequency [MHz] from:	3700 to: 4200
Nofitication reason:	○ Coordination ○ Notification ○ API ○ RES49 ◉ All

Select Data Items

date of protection
date of publication (IFIC)
date of receipt
earth station name
emission indicator
findings regulatory

>>

<<

class of station,
date of bringing into use,
date of receipt

Display Suppressed Stations: ○ Yes ◉ No

Display ... script: ○ Yes ◉ No

Submit Clear

GSO Window= 20 E ± 8 degrees

Frequency Band

Coordination between GSO/GSO under provision 9.7
Result of Exercise of Coordination Arc Approach

generated query										
satellite name	type	adm	ntwk org	category	longitude	freq_min	freq_max	stn cls	d inuse	d rcv
STATSIONAR-27	G	RUS		A	12	3400.00000	3950.00000	EC	01.06.1992	28.08.1987
STATSIONAR-27	G	RUS		C	12	3550.00000	3800.00000	EC	01.06.1992	23.08.2000
						3750.00000	3800.00000	EC	01.06.1992	23.08.2000
						3400.00000	3850.00000	EC	01.06.1992	23.08.2000
						3850.00000	3900.00000	EC	01.06.1992	23.08.2000
						3450.00000	3950.00000	EC	01.06.1992	23.08.2000
						3400.00000	3950.00000	EC	30.07.2001	23.08.2000
AST-12E	G	F		A	12	3400.00000	4200.00000	EC	01.06.2016	25.08.2009
EMARSAT-8W/M	G	UAE		A	12	3400.00000	4200.00000	EC	10.02.2015	18.10.2009
						3400.00000	4200.00000	EK	10.02.2015	18.10.2009
						3400.00000	4200.00000	ER	10.02.2015	18.10.2009
ASAT E012	G	F		A	12	3400.00000	4200.00000	EC	08.07.2017	23.07.2010
AZERSAT C12	G	AZE		A	12	3400.00000	4200.00000	EC	01.01.2016	19.11.2010
AST-2-12E	G	F		A	12	3400.00000	4200.00000	EC	01.01.2018	05.07.2011
AMS-B1-12E	G	ISR		A	12	3400.00000	4200.00000	EC	01.04.2018	16.02.2012
						3400.00000	4200.00000	EK	01.04.2018	16.02.2012
						3400.00000	4200.00000	ER	01.04.2018	16.02.2012
BASAT E012	G	F		A	12	3400.00000	4200.00000	EC	08.08.2018	01.03.2012
EMARSAT-9Y	G	UAE		A	12	3400.00000	4200.00000	EC	30.11.2018	30.11.2011
						3400.00000	4200.00000	EK	30.11.2018	30.11.2011
						3400.00000	4200.00000	ER	30.11.2018	30.11.2011
CHNSAT-12E	G	CHN		A	12	3400.00000	4200.00000	EC	01.01.2019	19.01.2012
AMS-B2-12E	G	ISR		A	12	3400.00000	4200.00000	EC	01.06.2019	17.06.2012
						3400.00000	4200.00000	EK	01.06.2019	17.06.2012
						3400.00000	4200.00000	ER	01.06.2019	17.06.2012
STATSIONAR-27	G	RUS		N	12	3400.00000	3950.00000	EC	23.08.2005	16.08.2005
IRANSAT-12.5E	G	IRN		A	12.5	3400.00000	4200.00000	EC	03.01.2019	03.01.2012
F-SAT-N-E-13E	G	F		C	13	3400.00000	4200.00000	EC	28.05.2016	01.12.2009
						3400.00000	4200.00000	EK	28.05.2016	01.12.2009
						3400.00000	4200.00000	ER	28.05.2016	01.12.2009
F-SAT-N-E-13E	G	F		A	13	3400.00000	4200.00000	EC	28.05.2016	28.05.2009
LUX-G7-4	G	LUX		A	13	3400.00000	4200.00000	EC	18.08.2016	19.08.2009

Coordination between GSO/GSO under provision 9.7

✓ Criterion of $\Delta T/T > 6\ \%$

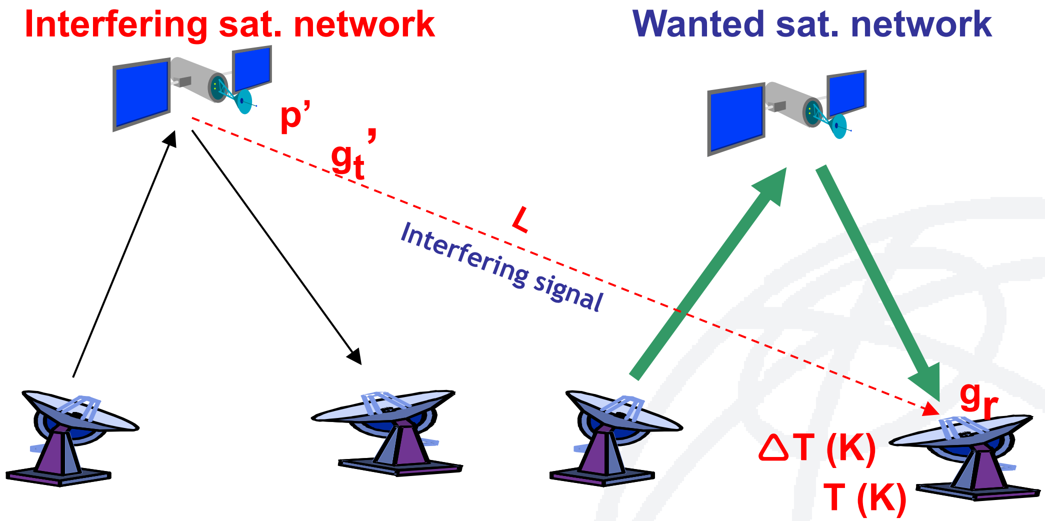
- Mask used by BR to establish coordination requirements in any other scenario where CA is not applicable.
- Utilized by Administrations to request BR to include or exclude networks in coordination process under No.9.41
- Described by Appendix 8 to RR.
- It measures the increase of Noise Temperature at Rx due to Interference as a generic method.
- It does not take into account: - wanted signal.
  - interfering spectrum shape

$\Delta T/T > 6\ \%$ => Potential Harmful Interference

Further detailed analysis is needed to ensure that coordination is really needed (e.g. C/I)

$\Delta T/T \leq 6\ \%$ => No Harmful Interference

ΔT/T : Introduction to General Method



AP8 describes the method including definitions

$$\Delta T / T = (p' \, g_t' \, g_r) / KLT$$

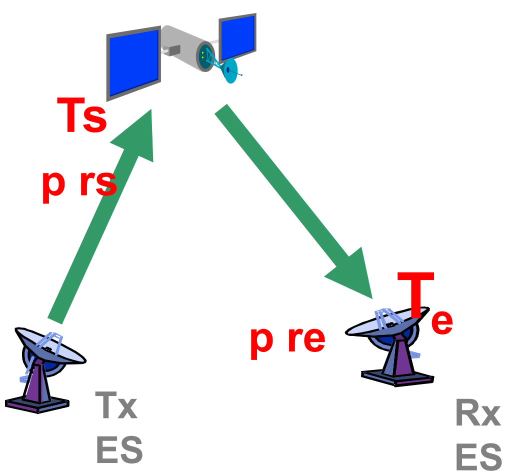
Interfering power density level

Transmission gain γ :

- ✓ Valid for Simple Freq. Changing Transponders (Bent Pipe) only.
- ✓ Not applicable when satellite has on-board signal processing (digital regenerating transponders, change of modulation, etc). This case requires separate treatment of up and downlinks.

$$\gamma = p_{re} / p_{rs}$$

Power received at the earth stn.
Power received at the satellite

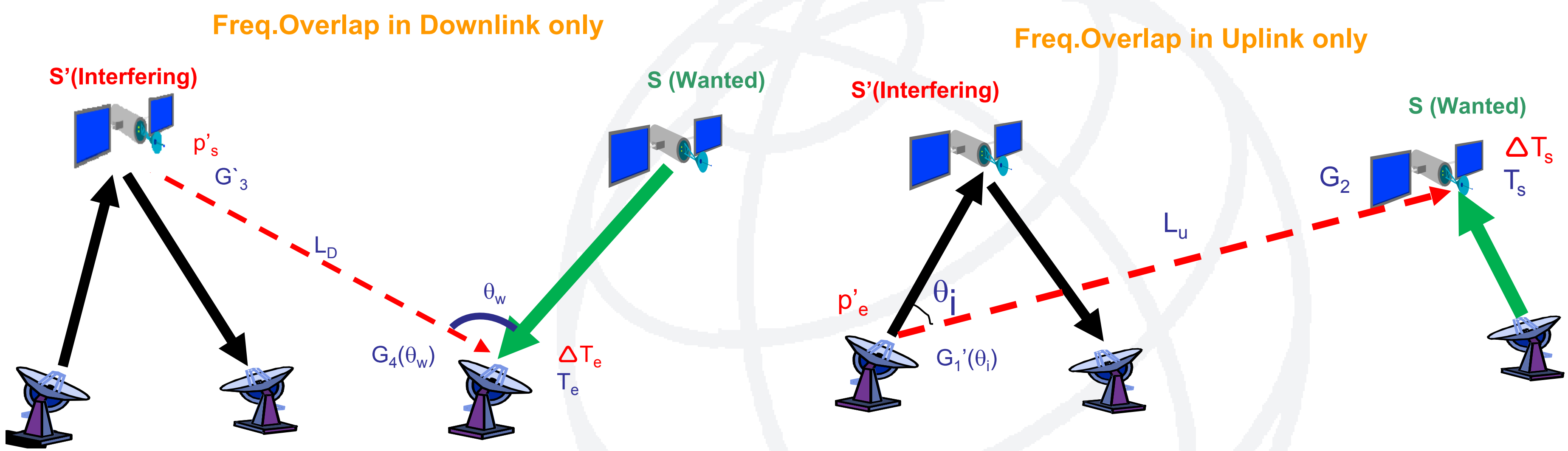


Equivalent Satellite Link Noise Temperature:

$$T = Te + \gamma \, Ts \ \ (K)$$

ΔT/T Case I : Freq. Overlap Co-Directional

Separate treatment of Up and Downlink
(Wanted Satellite has on-board signal processing)



$\Delta T / T = \Delta T_E / T_E$

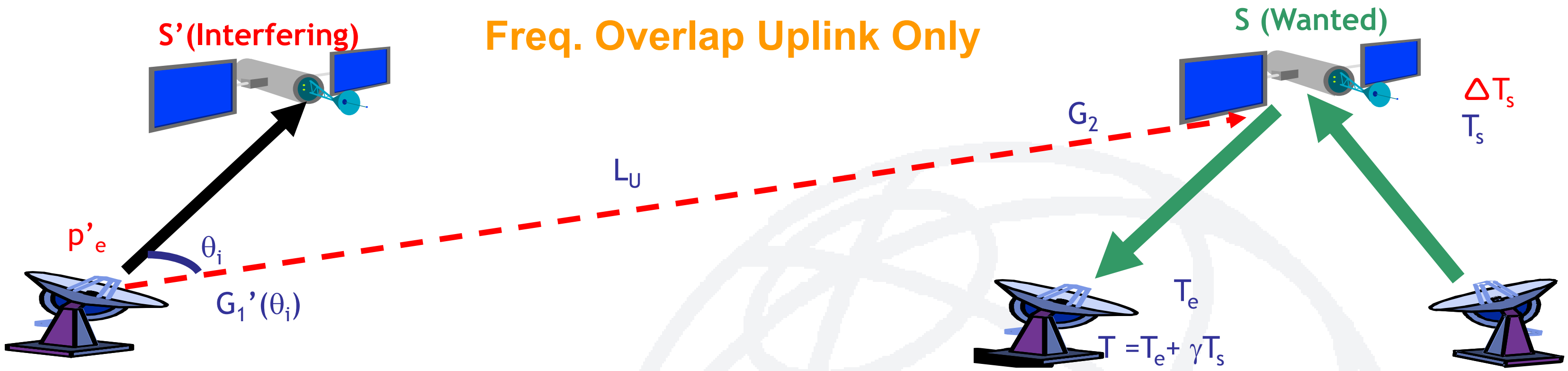
$\Delta T / T = 10\log(p'_s) + G`_3 - L_D + G_4(\theta_w) - K - T_E$ (dB)

$\Delta T / T = \Delta T_S / T_S$

$\Delta T / T = 10\log(p'e) + G`1(\theta i) - LU + G2 - K - TS$ (dB)

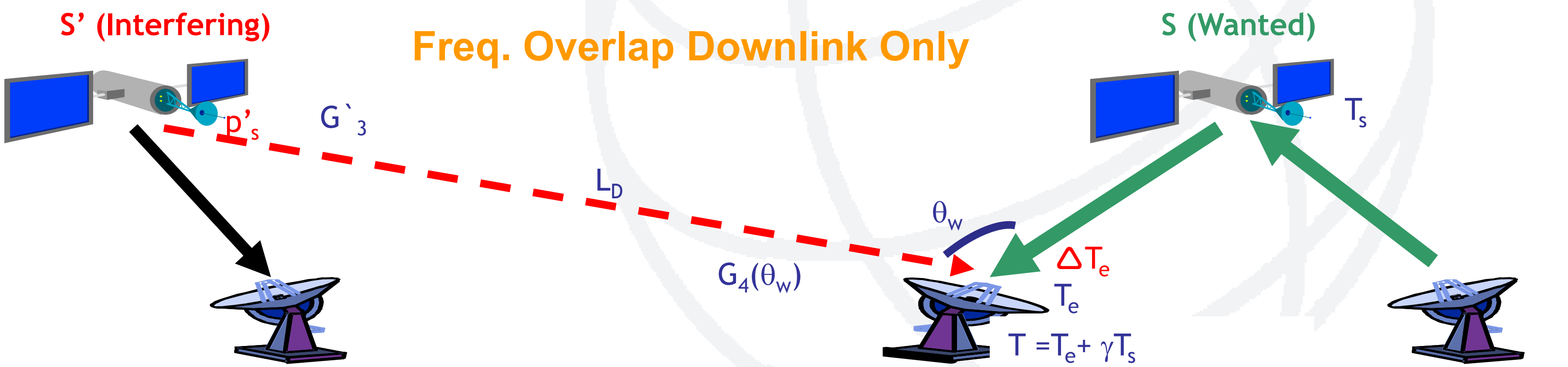
ΔT/T Case I : Freq. Overlap Co-Directional

Simple Freq.Changing Transponder (Bent Pipe)



$$\Delta T / T = \gamma \Delta T_s / T$$

$$\Delta T / T = 10\log \gamma + 10\log(p'_e) + G`_1(\theta_i) - L_U + G2 - K - T \text{ (dB)}$$

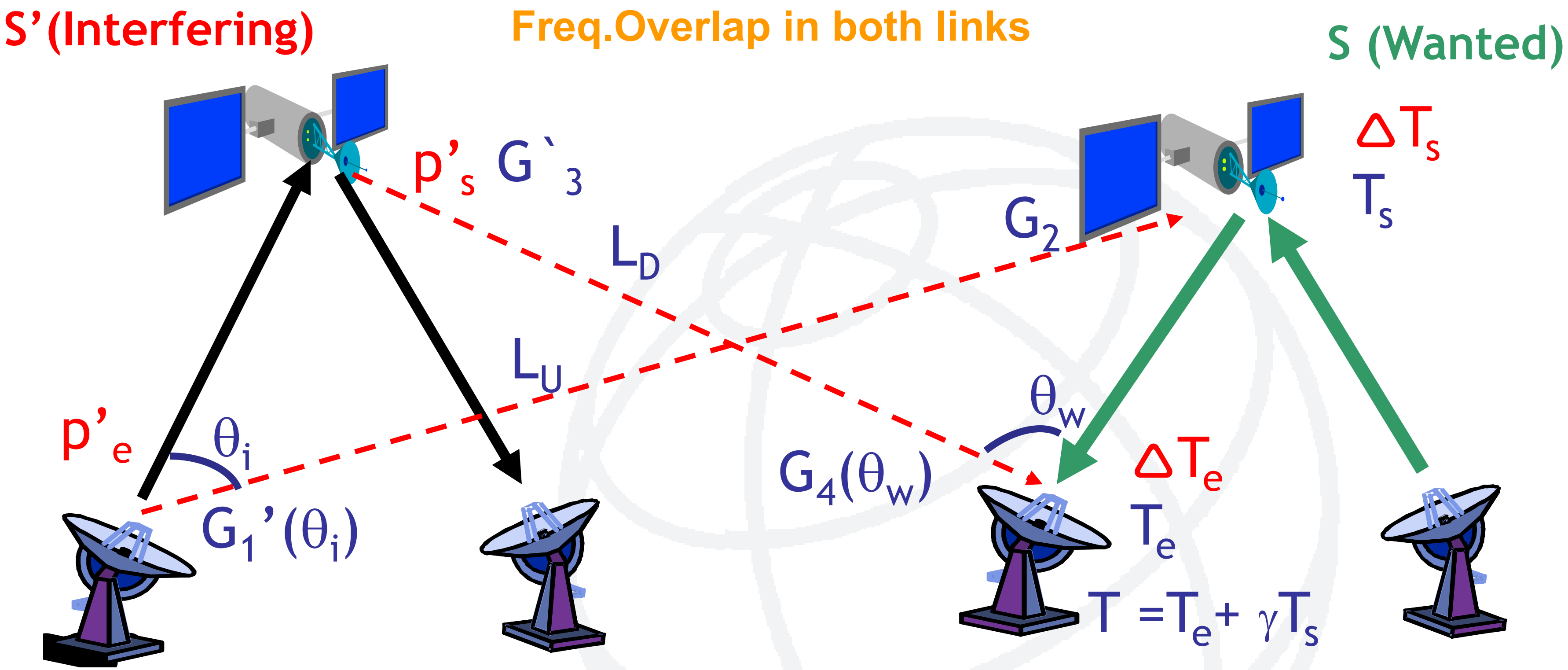


$$\Delta T / T = \Delta TE / T$$

$$\Delta T / T = 10\log(p'_s) + G`_3 - L_D + G4\,(\theta_W) - K - T \text{ (dB)}$$

ΔT/T Case I : Freq. Overlap Co-Directional

Simple Freq.Changing Transponder (Bent Pipe)



$$\Delta T/T = (\Delta T_e + \gamma\, \Delta T_s) / T$$

$$\Delta T / T = (p'_s\, g`_3\, g_{4(\theta_w)}) / (k\, l_D\, T) + \gamma\, (p'_e\, g`_{1(\theta_i)}\, g_2) / (k\, l_U\, T)$$

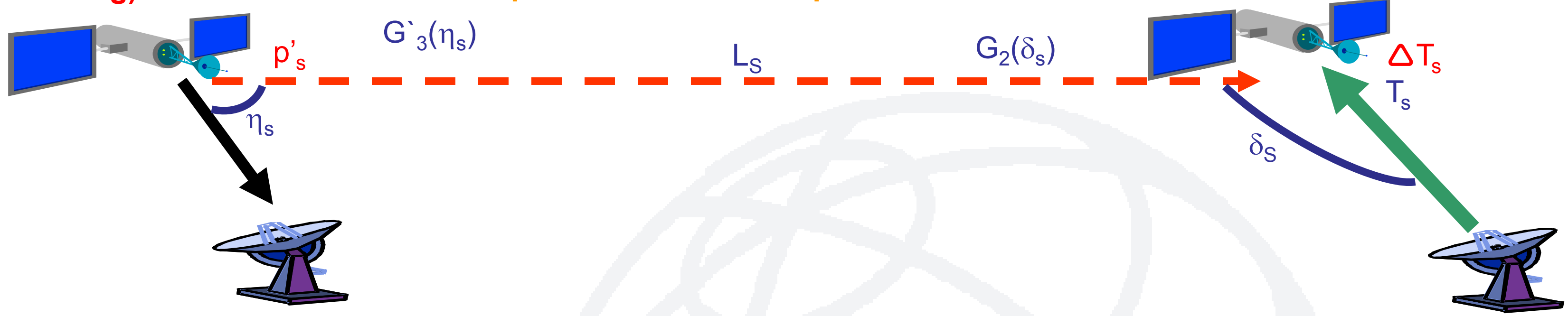
$\Delta T/T$ Case II: Freq.Overlap in Opposite Direction of Tx. (Inter-Satellite)

Downlink (interfering) overlaps Uplink(wanted)

S'(Interfering)

Separate treatment of up & down links

S (Wanted)



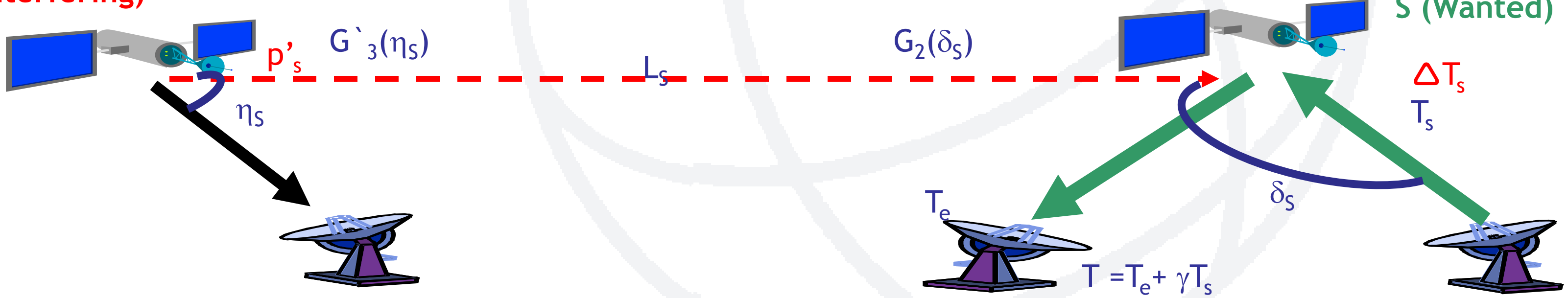
$\Delta T / T = \Delta T_S / T_S$

$\Delta T / T = 10\log(p'_s) + G`_3(\eta_S) - L_s + G_2(\delta_S) - K - T_s$ (dB)

Wanted Satellite has Simple Freq. Changing TXP (bent-pipe)

S'(Interfering)

S (Wanted)



$\Delta T/T = \gamma \Delta T_s / T$

$\Delta T/T = 10\log\gamma + 10\log(p'_s) + G`_3(\eta_S) - L_s + G_2(\delta_S) - K - T$(dB)

Coordination between GSO/GSO under provision 9.7

✓ C/I Criterion

- Utilized by BR to perform detailed examination of probability of harmful interference when so requested by Administrations under No.11.32A of RR.
- Based on methodology and protection criteria defined by REC ITU-R S.741-2 and associated Rule of Procedure from RRB, or by common agreement between Adms.
- It takes into account: - Wanted signal
  (level and type of carrier-modulation)
  - Interfering signal
  (level and spectrum shape)
  - Overlapped BandWidth
- More accurate to perform inter-networks sharing analysis, based on quality and availability objectives.
- Used by operators in coordination meetings.

Concept of C/I:

$$C/I = C/N + K$$

Protection ratio

(generally, between 12.2 - 14 dB, depending on the type of carriers)

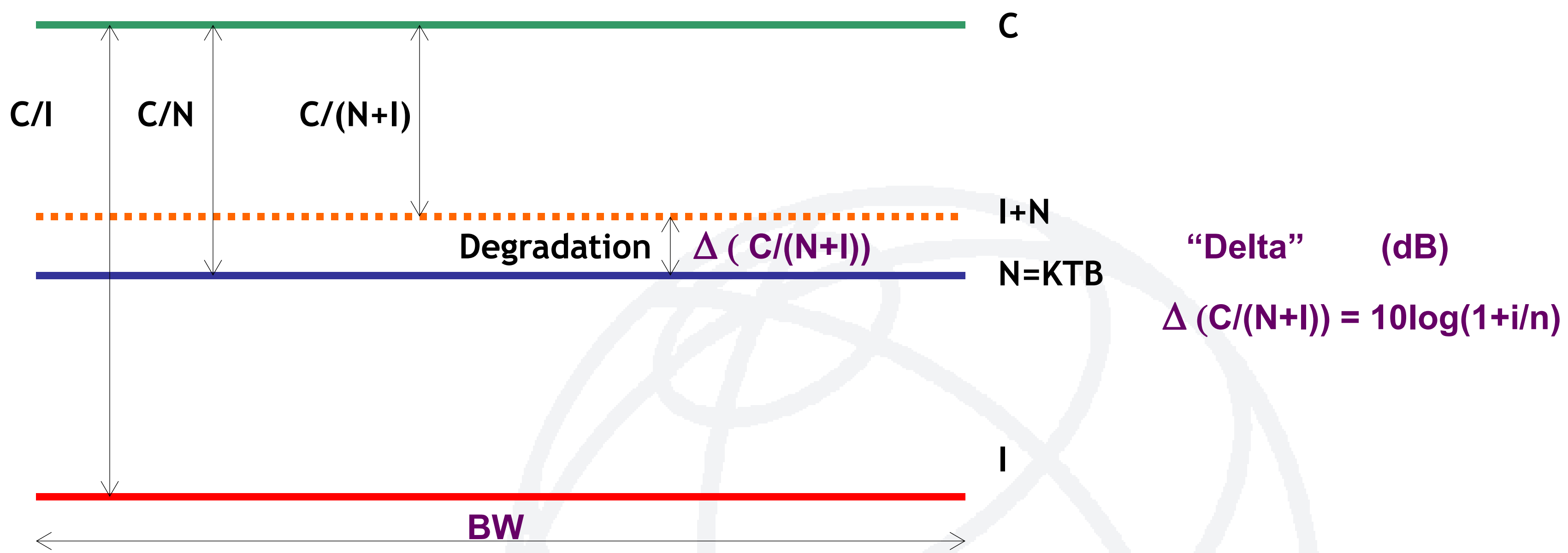
Result of your Link Budget

(considering objectives like S/N or BER, availability, etc)

Protection required to ensure compatibility between networks

****C/I examination will be presented in detail in a separate document****

Interference: Criteria and typical values



I/N=-12 dB → Degradation ≅ 0.26 dB → ΔT/T = 6%

I/N=-10 dB → 0.4 dB

I/N=- 6 dB → 1 dB

Multiple Interference Sources:

- Worst case given by

$$I_{Total} = 10.\log\left[\,10^{(I1/10)} + 10^{(I2/10)} + \ldots + 10^{(In/10)}\,\right]$$

where I_{Total}, I_1, $I_2 \ldots I_n$ are in dBW.

- In terms of C/I:

$$c/i_{Total} = \frac{1}{\dfrac{1}{c/i_{Adj.\ Sat.}} + \dfrac{1}{c/i_{Terrest}} + \dfrac{1}{c/i_{Other}}}$$

Under Study:

- **Resolution 756 (WRC-12), WRC-12 resolved to invite ITU-R:**

  1 to carry out studies to examine the **effectiveness and appropriateness** of the current criterion **(ΔT/T > 6%)** used in the application of **No. 9.41** and **consider any other possible alternatives** (including the alternatives outlined in Annexes 1 and 2 to this Resolution, such as **pfd masks or *C/I***, as appropriate), for the bands referred to in *recognizing e)*;

  2 to study whether **additional reductions in the coordination arcs** in RR Appendix 5 (Rev.WRC-12) are appropriate for the **6/4 GHz** and **14/10/11/12 GHz** frequency bands, and whether it is appropriate to reduce the coordination arc in the **30/20 GHz band.**

- **Director of BR will include the results in his Report to WRC-15.**

Ongoing studies are considering:

- which frequency bands would be subject to the new coordination criterion?
- new threshold would be based on single-entry or multiple sources of interference?
- to apply different criteria for each combination of interfering and interfered-with carrier type?
  (noting that Recommendation ITU-R S.741 may not take into account some modulation-coding schemes currently in use, as well as the difficulties of identifying them using the current RR Appendix 4 parameters)
- the possibility of defining a reasonable range of technical parameters (e.g. uplink G/T, downlink noise temperature, antenna sizes)
- identifying provisions and situations where the new criteria should be applied

Contributions to studies are based on:

- homogenous networks
- separation of two and three degrees
- possible levels of permissible interference and
- associated loss of energy margin and capacity

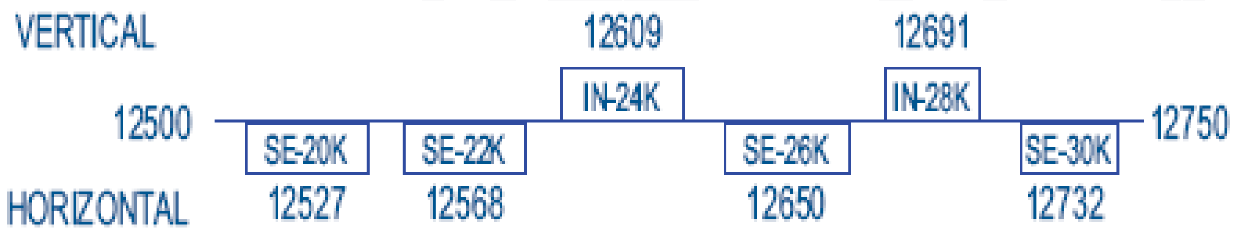
See Annex to the Working Party 4A Chairman’s Report

Methods to facilitate coordination and sharing scenario between GSO

- **Frequency Separation**
  - Band Segmentation
  - Channeling Plan
- **Polarization**
- **Improvement of antenna system spatial discrimination**
  - Design of Antenna gain contours, roll-off and service areas associated to satellite beams
  - Modifying antenna diameters in the ground segment
  - Improvement to Earth Station Radiation Pattern
- **To Adjust orbital separation between adjacent satellites.**
- **To Reorganize distribution of diferent types of carrier.**
- **Use of advanced modulation/FEC technologies (eg. DVB-S2), signal coding and processing techniques (spread spectrum or CDMA, etc).**
- **Re-engineering of the link budget, including modulation-FEC, power density levels, adjusting Quality and Availavibility Objectives in order to tolerate higher levels of interference.**

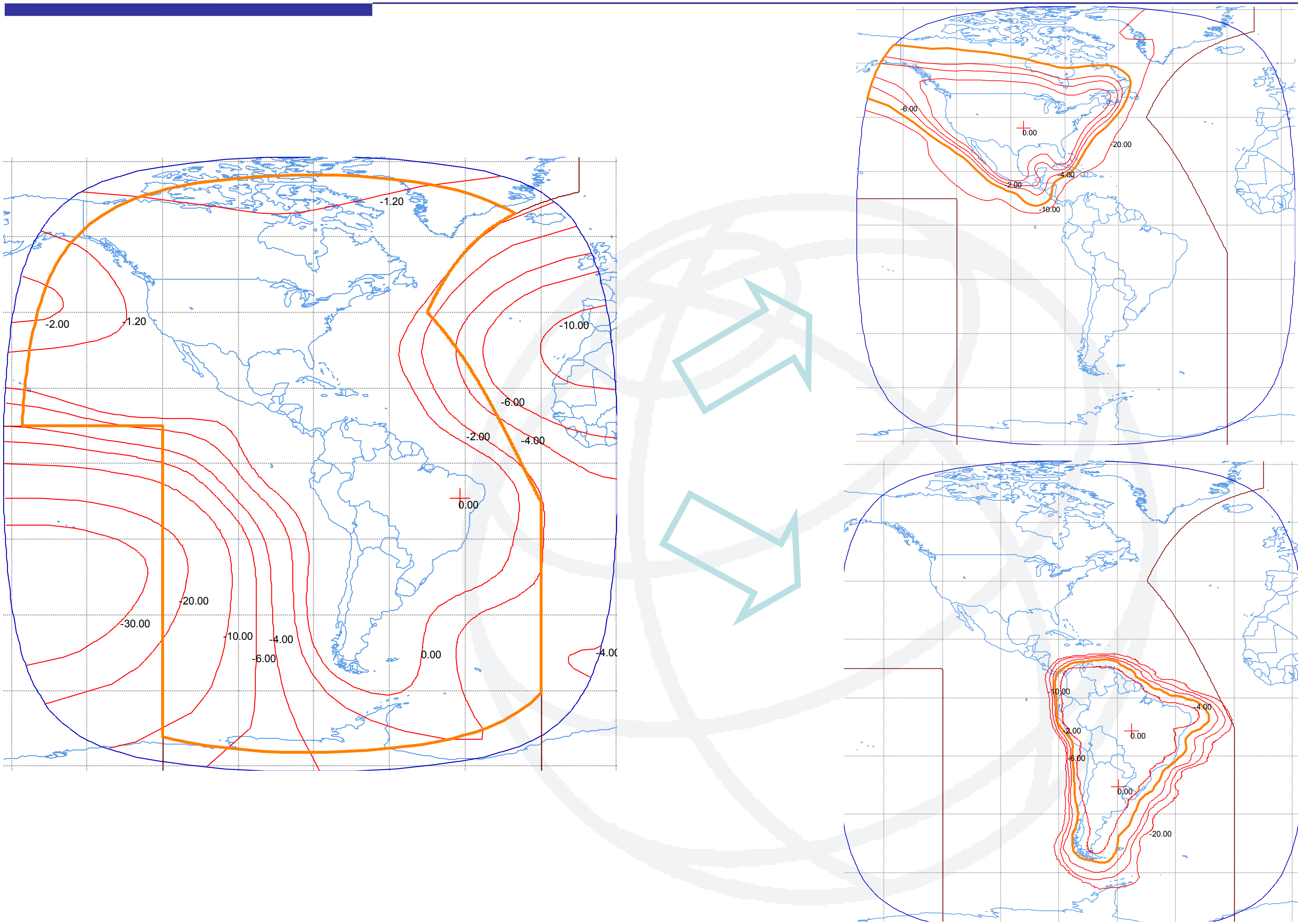
Frequencies - Polarizations

Satellite 1



Satellite 2

Space Segment - Spatial Discrimination



Modifying Orbital Separation:

Exercise:

Assuming $D/\lambda = 100$; ES Antenna Paterns REC 465-5 / REC 580-6

Interference Reduction:

$$I_f - I_i = 25.\log(\varphi_i / \varphi_f)$$

where φf: minimum final separation between satellites

φi: minimum initial separation between satellites

Scenario 1

$\Theta_{1n}-\Theta_{2n} = 2^\circ$ → Nominal Orbital Separation

$\Delta_{\Theta 1} = \Delta_{\Theta 2} = \pm 0.1^\circ$ → E-W Station Keeping

Interference Reduction with respect to Scenario 1

Scenario 2

$\Theta_{1n}-\Theta_{2n} = 3^\circ$

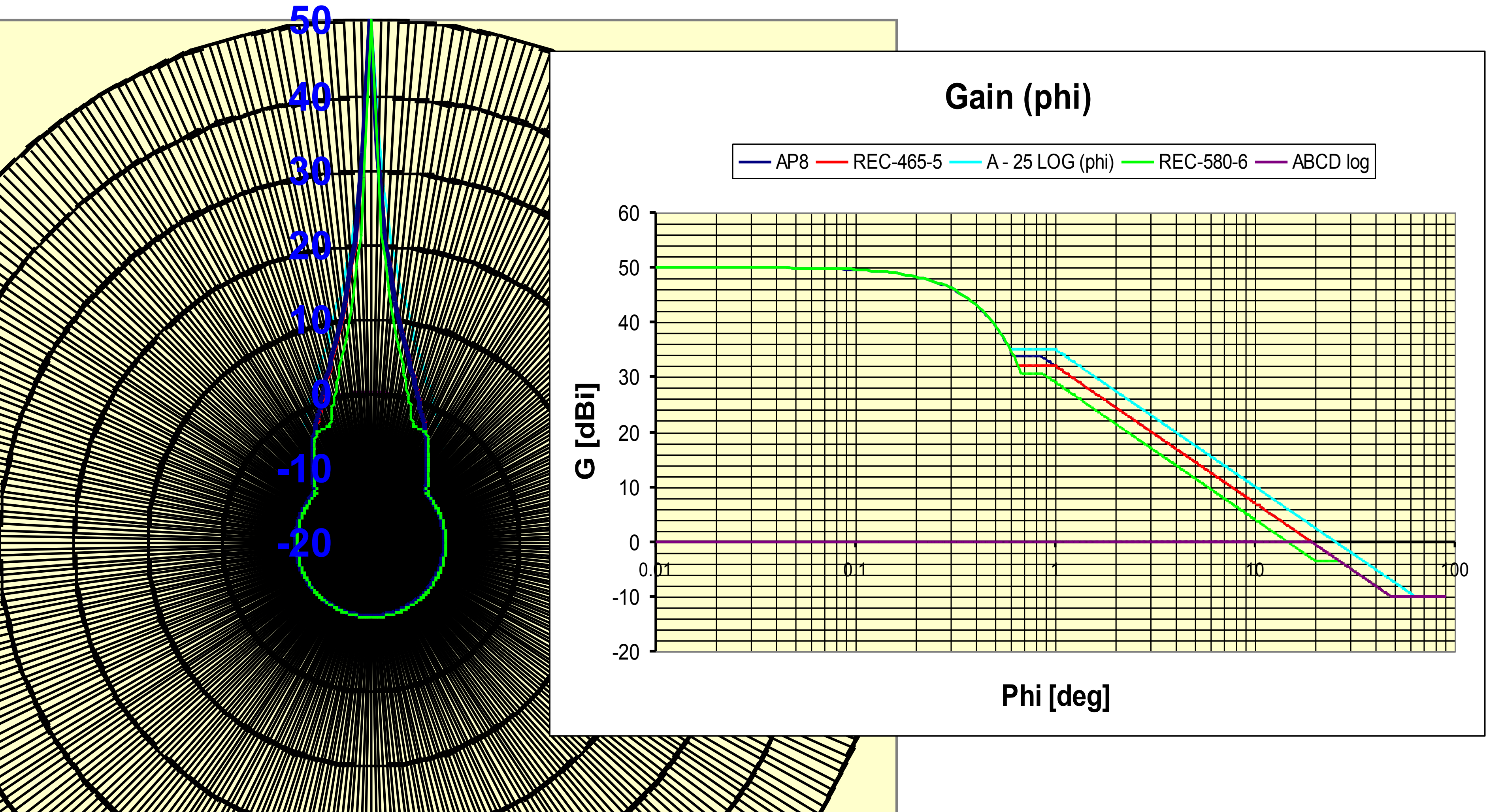
$\Delta_{\Theta 1} = \Delta_{\Theta 2} = \pm 0.1^\circ$

$I_f - I_i = 25.\log(1.8 / 2.8) =$ -4.8 dB

Warning: From Regulatory point of view, it may be an Impact of New Coordination Requirements in some cases due to increase of interference to other satellites

Ground Segment: Improving the Earth Station Radiation Pattern

Antenna Radiation Patterns Comparison
Sidelobes



Changing the Earth Station Antenna Diameter:

Ga max [dBi]=	43.2
Gb max [dBi]=	56

Mainlobe and Near-Sidelobes

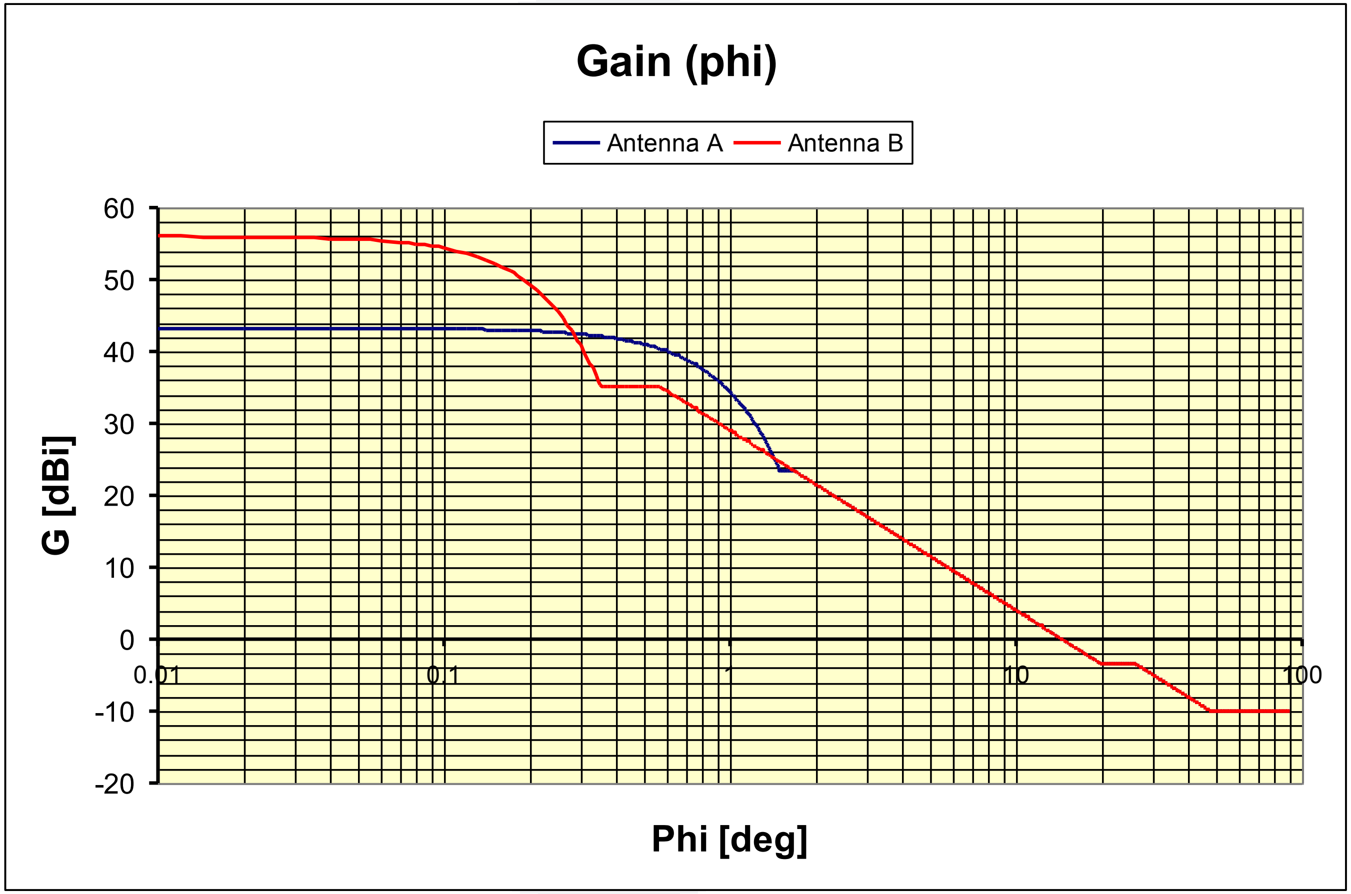
REC-580-6 Antenna Pattern

Antenna A

G1 =	23.34
Phi m =	1.50
D/L =	59.40
Phi r =	1.68
Phi b=	47.86
Beamwidth=	1.17

Antenna B

G1 =	35.21
Phi m =	0.35
D/L =	259.28
Phi r =	0.56
Phi b=	47.86
Beamwidth=	0.27



REFERENCES - COMMENTS:

Antenna A= Typical 1.2M

Antenna B= Typical 13M

Reorganizing distribution of different types of carrier

- **To identify diferent types of carriers such as:**
  - TT&C
  - Analog TV/FM
  - Digital Data

- **To consider their characteristics of diversity in terms of BW, Max. Power and spectral density distribution.**

- **To group them in the frequency domain taking into account the distribution of similar carriers used by neighboring satellites.**

- **Off-axis eirp masks associated to type of carriers and frequency bands, as well as operational restrictions or relaxations, may be agreed during the coordination process.**

How to Optimize a Filing to be submitted to ITU ?

Goal ? → Favorable Findings + Realistic Info ⇒ To Ensure Recording + Accuracy → MIFR

Positive Cycle for current & future networks

- Administrations are free to choose the way to organize a filing
- Coordination Request:
  - -needs certain flexibility of parameters
  - -may be a General Approach
- Notification:
  - -specific
  - -accurate parameters
  - -realistic

Filing is Group Structured

BR6a/BR6b Id. no. 103500259 — BR3a/BR3b Provision reference 11.2 N — BR2 Adm. serial no. — KR1 R

BR62 Expiry date for bringing into use — BR63 Confirmed date of bringing into use — BR64 Date of receipt of 1st Res49

BR14 Special Section

C4a Class of station: EC | EK | ED — C3a Assigned freq. band: 36000 — C5a Noise temperature: 600

C4b Nature of service: CO | CP | CV — C6a Polarization type: V — C6b Polarization angle

C11a1 Service area no.: 1 — C11a2 Service area — C11a3 Service area diagram: 2

A6/A5 Coordination/Agreements: 9.7 | O | | AUS USA/IT

C2a1 Assigned frequency									
14.02	GHz	14.1	GHz	14.18	GHz	14.26	GHz	14.34	GHz
14.06	GHz	14.14	GHz	14.22	GHz	14.3	GHz	14.38	GHz

A13 Ref. to Special Sections	
AR11/A	/1147
AR11/C	/3147

	C7a Design. of emission	C8a1/C8b1 Max. peak pwr	C8a2/C8b2 Max. pwr dens.	C8c1 Min. peak pwr	C8c2 Attch.	C8c3 Min. pwr dens.	C8c4 Attch.	C8e1 C/N ratio	C8e2 Attch.
1	14K0G7W--	-19.4	-60	-32.2		-72.8		6.8	
2	15M6G7W--	22	-49.2	5.3		-65.9		1.3	
3	36M0G7W--	18.4	-57.1	1.7		-73.8		5.1	
4	[illegible]	[illegible]	[illegible]	[illegible]		[illegible]		[illegible]	
5	5K00G1X--	-11.8	-47.5	-32.2		-68		8.5	
6	5M00G1X--	12.7	-51.2	-7.7		-71.7		4.8	
7	7M89G7W--	15.1	-53.2	-1.6		-69.9		0.9	

C10b1 Assoc. earth station id.	C10b2 Type	C10c1 Geographical coord.	C10c2 Ctry	C10d1/C10d2 Cls. / Nat.			C10d3 Max. iso. gain	C10d4 Bmwdth		C10d7 Ant. diameter	C8g1 Max. aggr. pwr.	C8g2 Aggr. bandwidth	C8g3 Transp. bandwidth = Aggr. bandwidth
STD D	T			1	TC	CO	52	0.39					
				2	TK	CP							
				3	TD	CV							
STD E	T			1	TC	CO	47.6	0.72					
				2	TK	CP							
				3	TD	CV							
STD F	T			1	TC	CO	45.1	1.41					
				2	TK	CP							
				3	TD	CV							
STD G	T			1	TC	CO	41.2	1.41					
				2	TK	CP							
				3	TD	CV							

C10b1 Assoc. earth station id.	C10d5a Co-polar antenna pattern: Co-polar ref. pattern	Coef. A	Coef. B	Coef. C	Coef. D	Phi1	Co-polar rad. diag.
STD F	REC-580						
STD D	REC-580						
STD E	REC-580						
STD G	REC-580						

13C Remarks

Diversity of: Beam/Service Areas — Frequencies — Emissions — Earth Stations

Reorganize Filing considering diversity of frequency assignments and respective progress in coordination

For example: To split groups of frequency assignments by

-Beam/Service Areas
- Group 11 → SA 1 → Coord. Completed → Fav. Finding → MIFR
- Group 12 → SA 2 → Coord. Not Completed → Further progress is required

-Frequencies (similar process)

-Emissions (similar process)

-Earth Stations
- Group 21 → Typical 9m → Coord. Completed → Fav. Finding → MIFR
- Group 22 → Typical 1.2m → Coord.Not Completed → Further progress is required

Remarks: locating worst cases in separated groups will ensure recording of successfully coordinated frequency assignments.

References:

Some Useful Recommendations:

ITU-R S.741-2	Carrier-to-interference calculations between networks in the FSS.
ITU-R S.740	Technical coordination methods for fixed-satellite networks
ITU-R SM.1132	General principles and methods for sharing between radiocommunication services or between radio stations
ITU-R S.738	Procedure for determining if coordination is required between geostationary-satellite networks sharing the same frequency bands
ITU-R S.580-6	Radiation diagrams for use as design objectives for antennas of earth stations operating with geostationary satellites
ITU-R S.465-5	Reference radiation pattern of earth station antennas in the fixed-satellite service for use in coordination and interference assessment in the frequency range from 2 to 31 GHz
ITU-R S.1855	Alternative reference radiation pattern for earth station antennas used with satellites in the geostationary-satellite orbit for use in coordination and/or interference assessment in the frequency range from 2 to 31GHz

References:

ITU-R S.524-9 Maximum permissible levels of off-axis e.i.r.p. density from earth stations in geostationary-satellite orbit networks operating in the fixed-satellite service transmitting in the 6 GHz, 13 GHz, 14 GHz and 30 GHz frequency bands

ITU-R S.1328-4 Satellite system characteristics to be considered in frequency sharing analyses within the fixed-satellite service

Appendix 8 to Radio Regulations (Volume 2 of RR)

Rules of Procedure of Radio Regulations Board associated to C/I analysis under 11.32A

Handbook on Satellite Communications – ITU, Wiley